DAYANANDA SAGAR COLLEGE OFARTS SCIENCE AND COMMERCE

Bachelor of Computer Applications

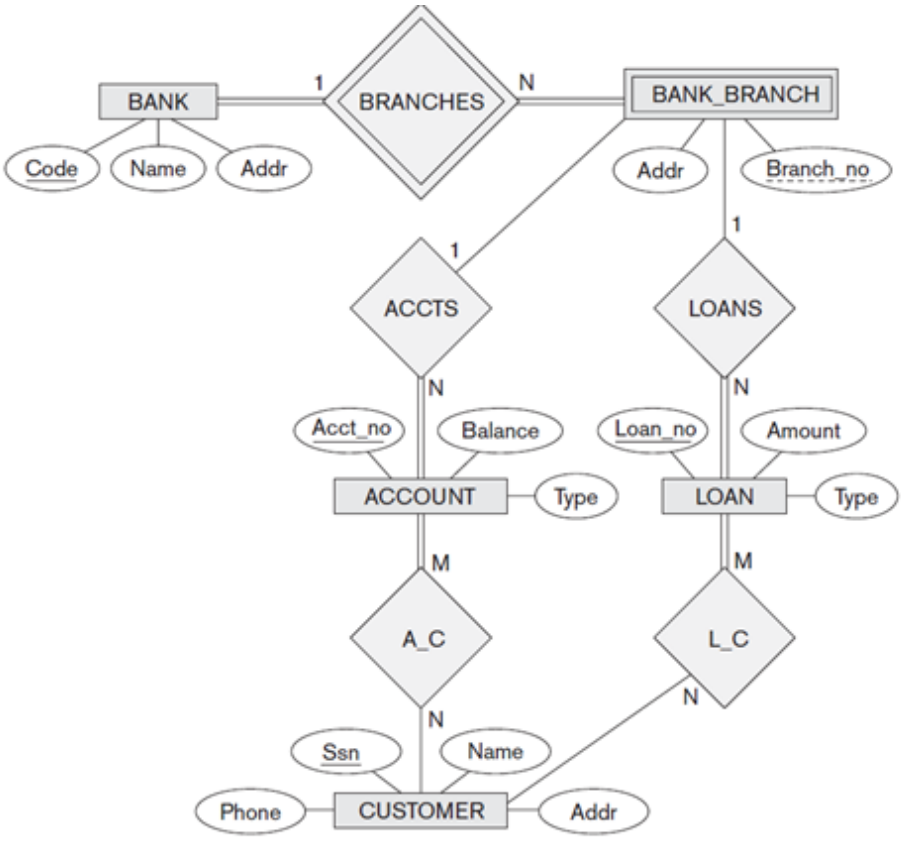
CA- C10P: DATABASE MANAGEMENT SYSTEMS

LAB MANUAL

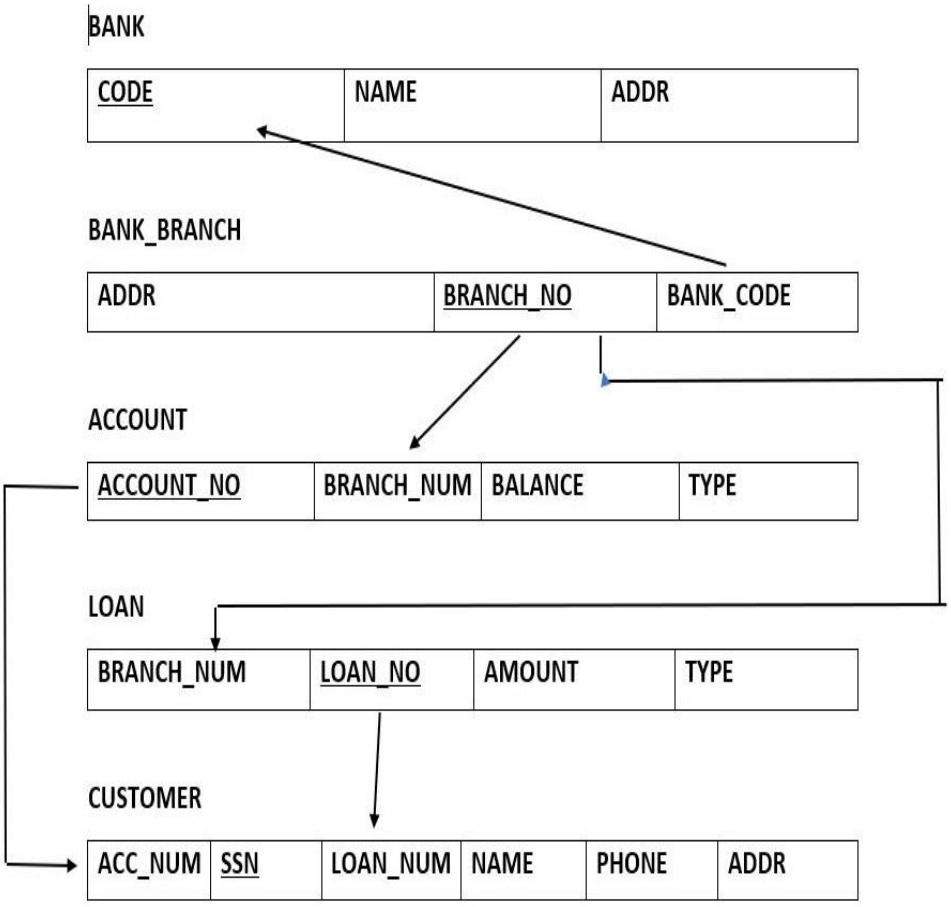
PART – A

1.a Draw E-R diagram for Bank database and convert entities and relationships to relation table for a given scenario.

E-R DIAGRAM FOR BANK DATABASE

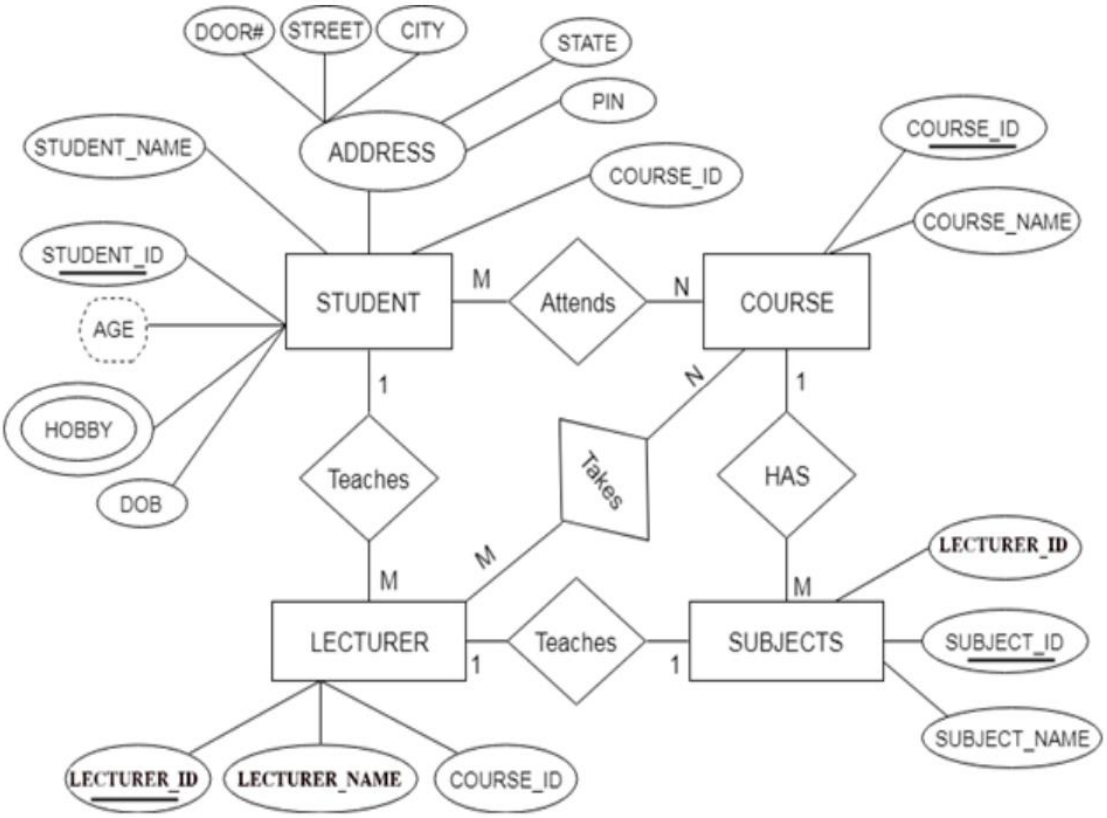


RELATIONAL SCHEMA FOR BANK DATABASE

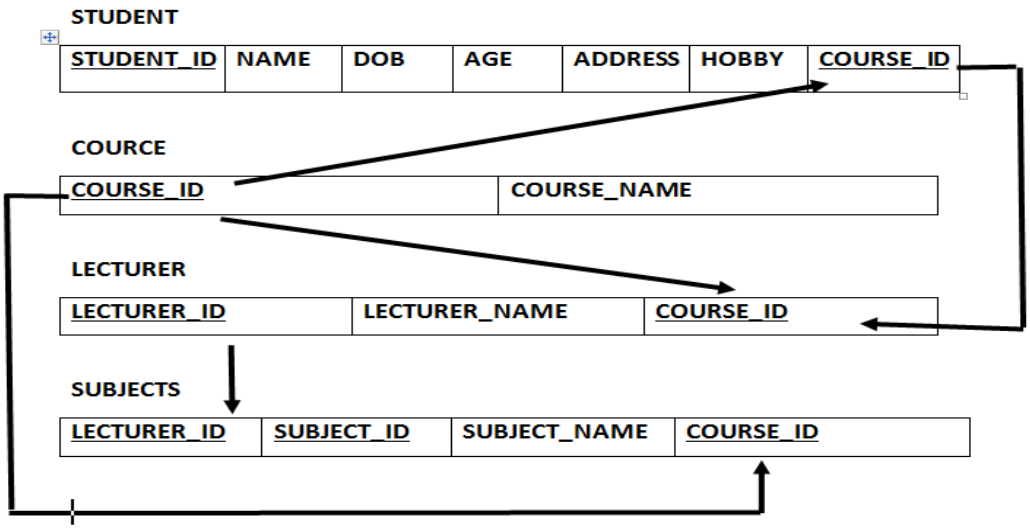


1.b Draw E-R diagram for College and convert entities and relationships to relation table for a given scenario.

E-R DIAGRAM FOR COLLEGE



RELATIONAL SCHEMA FOR COLLEGE DATABASE



2. **Consider the Company database with following Schema . Draw E-R diagram and convert entities and relationships to relation table for a given scenario.**

COMPANY DATABASE SCHEMA :

EMPLOYEE (FNAME, MINIT, LNAME, SSN, BDATE, ADDRESS, SEX, SALARY, SUPERSSN,DNO)

DEPARTMENT (DNAME, DNUMBER,MGRSSN, MGRSTARTDATE)

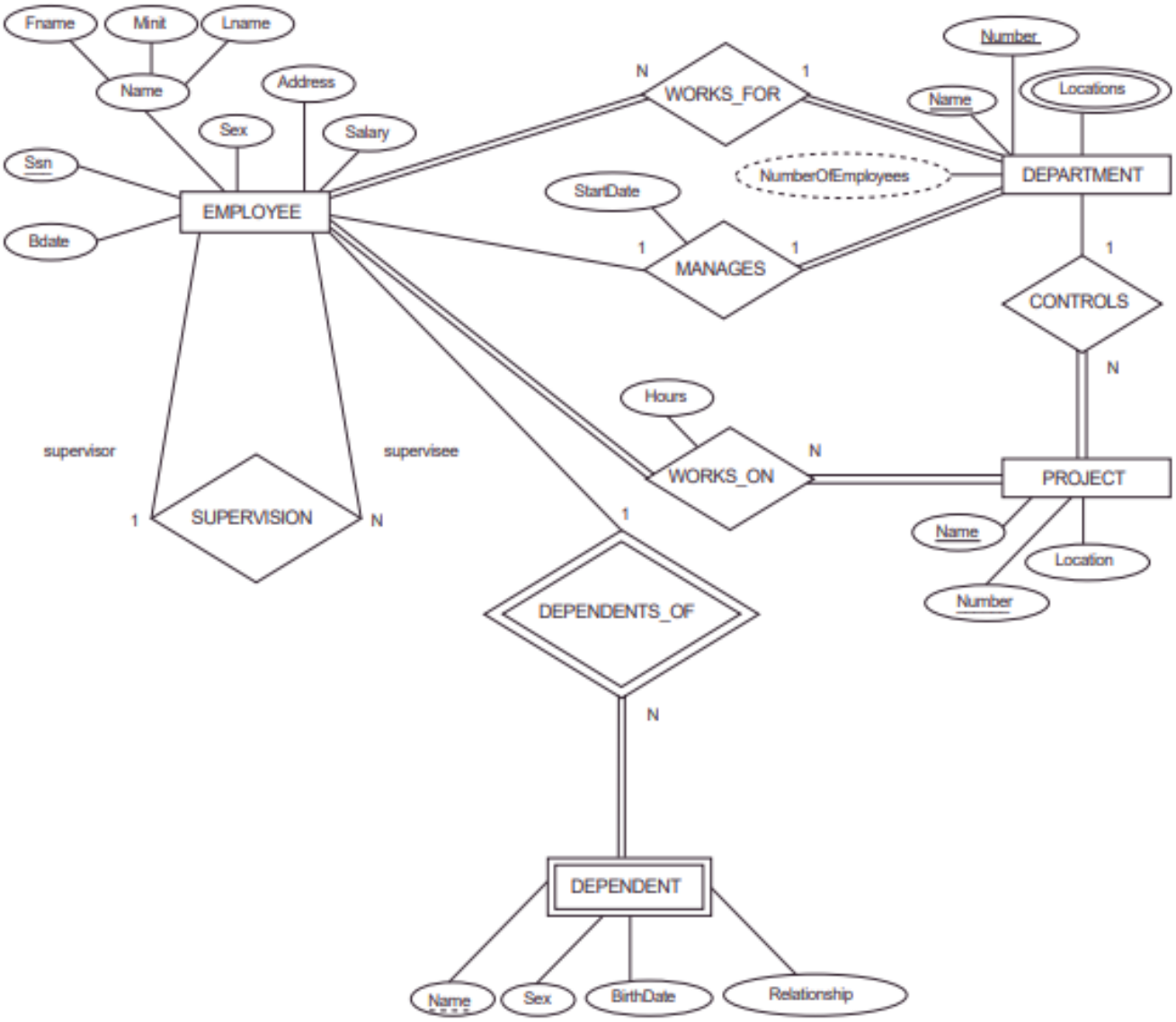
DEPT_LOCATIONS (DNUMBER, DLOCATION)

PROJECT (PNAME, PNUMBER, PLOCATION, DNUM)

WORKS_ON (ESSN, PNO,HOURS)

DEPENDENT (ESSN, DEPENDENT_NAME, SEX, BDATE, RELATIONSHIP)

DRAW E-R DIAGRAM FOR COMPANY DATABASE:



Viewing all tables in a database:

SQL> select * from tab;

3.a Creating Tables

Creation of Employee table:

SQL> create table employee (fname varchar(15) not null, minit char(1), lname varchar(15) not null, ssn varchar (10) primary key, bdate date, address varchar(60), sex char(1), salary number(10,2), super_ssn varchar(10),dno number(10) not null);

Output:

```
SQL> create table employee
  2      (fname varchar(15) not null,
  3      minit char(1),
  4      lname varchar(15) not null,
  5      ssn varchar (10) primary key,
  6      bdate date,
  7      address varchar(60),
  8      sex char(1),
  9      salary number(10,2),
 10      super_ssn varchar(10),
 11      dno  number (10) not null  );

Table created.
```

SQL>desc employee;

Output:

```
SQL> desc employee;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 FNAME                                     NOT NULL VARCHAR2(15)
 MINIT                                              CHAR(1)
 LNAME                                     NOT NULL VARCHAR2(15)
 SSN                                       NOT NULL VARCHAR2(10)
 BDATE                                              DATE
 ADDRESS                                            VARCHAR2(60)
 SEX                                                CHAR(1)
 SALARY                                             NUMBER(10,2)
 SUPER_SSN                                          VARCHAR2(10)
 DNO                                       NOT NULL NUMBER(10)
```

Creation of Department table:

SQL>create table department (dname varchar(25) not null, dnumber number (10) primary key, mgrssn varchar(10) not null , mgrstartdate date,foreign key(mgrssn) references employee(ssn));

Output:

```
SQL> create table department
  2  (dname varchar(25) not null,
  3      dnumber  number (10)  primary key,
  4      mgrssn  varchar(10) not null,
  5      mgrstartdate date,
  6     foreign key(mgrssn) references employee(ssn));

Table created.
```

SQL>desc department;

Output:

```
SQL> desc department;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DNAME                                     NOT NULL VARCHAR2(25)
 DNUMBER                                   NOT NULL NUMBER(10)
 MGRSSN                                    NOT NULL VARCHAR2(10)
 MGRSTARTDATE                                       DATE
```

Creation of dept_locations table:

SQL>create table dept_locations (dnumber number(10) not null, dlocation varchar(25) not null, foreign key(dnumber) references department(dnumber));

SQL>desc dept_locations;

Output:

```
SQL> create table dept_locations
  2      (dnumber  number(10) not null,
  3      dlocation varchar(25) not null,
  4      foreign key(dnumber) references department(dnumber));

Table created.

SQL> desc dept_locations;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DNUMBER                                   NOT NULL NUMBER(10)
 DLOCATION                                 NOT NULL VARCHAR2(25)
```

Creation of Project table:

SQL> create table project (pname varchar(15) not null, pnumber number(10) primary key , plocation varchar(15), dnum number (10) ,foreign key(dnum) references department(dnumber));

SQL>desc project;

Output:

```
SQL> create table project
  2       (pname varchar(15) not null,
  3        pnumber number(10)  primary key ,
  4        plocation varchar(15),
  5        dnum  number (10) ,
  6         foreign key(dnum) references department(dnumber));

Table created.

SQL> desc project;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 PNAME                                     NOT NULL VARCHAR2(15)
 PNUMBER                                   NOT NULL NUMBER(10)
 PLOCATION                                          VARCHAR2(15)
 DNUM                                               NUMBER(10)
```

Creation of Works_on table:

SQL>create table works_on (essn varchar(10) not null , pno number (10) not null , hours number (3,1), foreign key(essn) references employee(ssn), foreign key(pno) references project(pnumber));

SQL>desc works_on;

Output:

```
SQL> create table works_on(essn  varchar(10)  not null ,pno number (10)  not null ,hours number (3,1) ,foreign key(essn)
 references employee(ssn),foreign key(pno) references project(pnumber));

Table created.

SQL> desc works_on
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ESSN                                      NOT NULL VARCHAR2(10)
 PNO                                       NOT NULL NUMBER(10)
 HOURS                                              NUMBER(3,1)
```

Creation of dependent table:

SQL>create table dependent (essn varchar (10) ,dependentname varchar(35) , sex char(1), bdate date,relationship varchar(25), foreign key(essn) references employee(ssn));

Output:

```
SQL> create table dependent
  2       (essn varchar (10)  ,
  3       dependentname  varchar(35) ,
  4       sex char(1),
  5       bdate date,
  6       relationship varchar(25),
  7         foreign key(essn) references employee(ssn));

Table created.
```

SQL>desc dependent;

Output:

```
SQL> desc  dependent;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ESSN                                               VARCHAR2(10)
 DEPENDENTNAME                                      VARCHAR2(35)
 SEX                                                CHAR(1)
 BDATE                                              DATE
 RELATIONSHIP                                       VARCHAR2(25)
```

Inserting/Updating/Deleting

Inserting

b. Inserting into employee table:

SQL>Insert into employee values ('darshan','B','anandh','e01', '09-Jan-1965'
,'731 Fondren, Houston, TX','M',30000, 'e02',5);

SQL> Insert into employee values ('mohan','T','ram','e02', '08-Dec-1955','638

Voss, Houston, TX','M',40000,'e08',5);

SQL>Insert into employee values ('Anisha','J','Bhatt','e03','19-Jan-1968','3321
Castle, Spring, TX','F',25000, 'e04',4);

SQL>Insert into employee values ('samantha','S','josh','e04','20-Jun-1941','291
Berry, Bellaire, TX','F',43000, 'e08' ,4);

SQL>Insert into employee values ('Ramesh','K','Narayan','e05','15-Sep-1962
','975 Fire Oak, Humble, TX','M',38000, 'e02' ,5);

SQL>Insert into employee values ('lisa','A','roy','e06','31-July-1972','5631
Rice ,Houston,TX','F',25000, 'e02',5);

SQL>Insert into employee values ('Ahmed','V','Jabbar','e07','29-Mar-1969','980
Dallas, Houston, TX','M',25000, 'e03',4);

SQL>Insert into employee values ('Jayanth','E','prasad','e08','10-Nov-1937','450 Stone, Houston, TX','M',55000,NULL,1);

SQL>select * from employee;

Output :

```
SQL> select * from employee;

FNAME           M LNAME         SSN       BDATE     ADDRESS                                                  S    SALARY SUPER_SSN        DNO
--------------- - --------------- --------- --------- -------------------------------------------------------- - ---------- --------- ----------
darshan         B anandh        e01       09-JAN-65 731 Fondren, Houston, TX                                 M     30000 e02              5
mohan           T ram           e02       08-DEC-55 638 Voss, Houston, TX                                    M     40000 e08              5
Anisha          J Bhatt         e03       19-JAN-68 3321 Castle, Spring, TX                                  F     25000 e04              4
samantha        S josh          e04       20-JUN-41 291 Berry, Bellaire, TX                                  F     43000 e08              4
Ramesh          K Narayan       e05       15-SEP-62 975 Fire Oak, Humble, TX                                 M     38000 e02              5
lisa            A roy           e06       31-JUL-72 5631 Rice ,Houston,TX                                    F     25000 e02              5
Ahmed           V Jabbar        e07       29-MAR-69 980 Dallas, Houston, TX                                  M     25000 e03              4
Jayanth         E prasad        e08       10-NOV-37 450 Stone, Houston, TX                                   M     55000                  1
```

SQL>insert into employee(fname,minit,lname,ssn,bdate,address,sex,salary,super_ssn,dno)values('TOM','Z','HANKS', 'e09','14-feb-1971','23 , Springton, TeXas','M',38000,null,1);

Output :

```
SQL> select * from employee;

FNAME           M LNAME         SSN       BDATE     ADDRESS                                                  S    SALARY SUPER_SSN        DNO
--------------- - --------------- --------- --------- -------------------------------------------------------- - ---------- --------- ----------
darshan         B anandh        e01       09-JAN-65 731 Fondren, Houston, TX                                 M     30000 e02              5
mohan           T ram           e02       08-DEC-55 638 Voss, Houston, TX                                    M     40000 e08              5
Anisha          J Bhatt         e03       19-JAN-68 3321 Castle, Spring, TX                                  F     25000 e04              4
samantha        S josh          e04       20-JUN-41 291 Berry, Bellaire, TX                                  F     43000 e08              4
Ramesh          K Narayan       e05       15-SEP-62 975 Fire Oak, Humble, TX                                 M     38000 e02              5
lisa            A roy           e06       31-JUL-72 5631 Rice ,Houston,TX                                    F     25000 e02              5
Ahmed           V Jabbar        e07       29-MAR-69 980 Dallas, Houston, TX                                  M     25000 e03              4
Jayanth         E prasad        e08       10-NOV-37 450 Stone, Houston, TX                                   M     55000                  1
TOM             Z HANKS         e09       14-FEB-71 23 , Springton, TeXas                                    M     38000                  1

9 rows selected.
```

---

ii. Updating :

SQL>update employee set salary =50000 where ssn='e01';

SQL>select * from employee;

Output :

```
SQL> select * from employee;

FNAME           M LNAME          SSN       BDATE     ADDRESS                                                          S     SALARY SUPER_SSN        DNO
--------------- - ---------------- --------- --------- ---------------------------------------------------------------- - ---------- ---------- ----------
darshan         B anandh         e01       09-JAN-65 731 Fondren, Houston, TX                                         M      50000 e02               5
mohan           T ram            e02       08-DEC-55 638 Voss, Houston, TX                                            M      40000 e08               5
Anisha          J Bhatt          e03       19-JAN-68 3321 Castle, Spring, TX                                          F      25000 e04               4
samantha        S josh           e04       20-JUN-41 291 Berry, Bellaire, TX                                          F      43000 e08               4
Ramesh          K Narayan        e05       15-SEP-62 975 Fire Oak, Humble, TX                                         M      38000 e02               5
lisa            A roy            e06       31-JUL-72 5631 Rice ,Houston,TX                                            F      25000 e02               5
Ahmed           V Jabbar         e07       29-MAR-69 980 Dallas, Houston, TX                                          M      25000 e03               4
Jayanth         E prasad         e08       10-NOV-37 450 Stone, Houston, TX                                           M      55000                   1
TOM             Z HANKS          e09       14-FEB-71 23 , Springton, TeXas                                            M      38000                   1

9 rows selected.
```

c. Deleting :

SQL>select * from employee where ssn='e08';

Output :

```
SQL> select * from employee where  ssn='e08';

FNAME           M LNAME          SSN       BDATE     ADDRESS                                                          S     SALARY SUPER_SSN        DNO
--------------- - ---------------- --------- --------- ---------------------------------------------------------------- - ---------- ---------- ----------
jayanth         E prasad         e08       10-NOV-37 450 Stone, Houston, TX                                           M      55000                   1
```

SQL>delete from employee where ssn='e08';

SQL>select * from employee;

Output :

```
SQL> select * from employee;

FNAME           M LNAME          SSN       BDATE     ADDRESS                                                          S     SALARY SUPER_SSN        DNO
--------------- - ---------------- --------- --------- ---------------------------------------------------------------- - ---------- ---------- ----------
darshan         B anandh         e01       09-JAN-65 731 Fondren, Houston, TX                                         M      50000 e02               5
mohan           T ram            e02       08-DEC-55 638 Voss, Houston, TX                                            M      40000 e08               5
Anisha          J Bhatt          e03       19-JAN-68 3321 Castle, Spring, TX                                          F      25000 e04               4
samantha        S josh           e04       20-JUN-41 291 Berry, Bellaire, TX                                          F      43000 e08               4
Ramesh          K Narayan        e05       15-SEP-62 975 Fire Oak, Humble, TX                                         M      38000 e02               5
lisa            A roy            e06       31-JUL-72 5631 Rice ,Houston,TX                                            F      25000 e02               5
Ahmed           V Jabbar         e07       29-MAR-69 980 Dallas, Houston, TX                                          M      25000 e03               4
TOM             Z HANKS          e09       14-FEB-71 23 , Springton, TeXas                                            M      38000                   1

8 rows selected.
```

SQL>Inserting into department values
SQL>Insert into department values ('Administration',4, 'e04' ,'01-Jan-1995');

SQL>Insert into department values ('Headquarters',1, 'e06',' 19-Jun-1981');

Output :

```
SQL> select * from department;

DNAME                          DNUMBER MGRSSN     MGRSTARTD
------------------------------ ---------- ---------- ---------
Research                            5 e02        22-MAY-88
Administration                      4 e04        01-JAN-95
Headquarters                        1 e06        19-JUN-81
```

Inserting into dept_locations table:

SQL>insert into dept_locations values (1,'Houston');

SQL>insert into dept_locations values (4,'Stafford');

SQL>insert into dept_locations values (5,'Bellaire');

SQL>insert into dept_locations values (5,'Sugarland');

SQL>insert into dept_locations values (5,'Houston');

SQL>select * from dept_locations;

Output :

```
SQL> select * from dept_locations;

   DNUMBER DLOCATION
---------- ------------------------------
         1 Houston
         4 Stafford
         5 Bellaire
         5 Sugarland
         5 Houston
```

Inserting into project table:

SQL>insert into project values('ProductX',1,'Bellaire',5);

SQL>insert into project values('ProductY',2,'Sugarland',5);

SQL>insert into project values('ProductZ',3,'Houston',5);

SQL>insert into project values('computerization',10,'stafford',4);

SQL>insert into project values('Reorganization',20,'Houston',1);

SQL>insert into project values('Newbenefits',30,'Stafford',4);

SQL>select * from project;

Output :

```
SQL> select * from project;

PNAME              PNUMBER PLOCATION             DNUM
--------------- ---------- --------------- ----------
ProductX                 1 Bellaire                 5
ProductY                 2 Sugarland                5
ProductZ                 3 Houston                  5
Computerization         10 Stafford                 4
Reorganization          20 Houston                  1
Newbenefits             30 Stafford                 4

6 rows selected.
```

Inserting into works_on table:

SQL>insert into works_on values(‘e01’,1,32.5);

SQL>insert into works_on values(‘e01’,2,7.5);

SQL>insert into works_on values(‘e05’,3,40.0);

SQL>insert into works_on values(‘e06’,1,20.0);

SQL>insert into works_on values(‘e06’,2,20.0);

SQL>insert into works_on values(‘e02’,2,10.0);

SQL>insert into works_on values(‘e02’,3,10.0);

SQL>insert into works_on values(‘e02’,10,10.0);

SQL>insert into works_on values(‘e02’,20,10.0);

SQL>insert into works_on values(‘e03’,30,30.0);

SQL>insert into works_on values(‘e03’,10,10.0);

SQL>insert into works_on values(‘e07’,10,35.0);

SQL>insert into works_on values(‘e07’,30,5.0);

SQL>insert into works_on values(‘e04’,30,20.0);

SQL>insert into works_on values(‘e04’,20,15.0);

SQL>insert into works_on values(‘e09’,20,NULL);

SQL>select * from works_on;

Output :

```
SQL> Select * from works_on;

ESSN                PNO       HOURS
---------- ---------- ----------
e01                  1        32.5
e01                  2         7.5
e05                  3          40
e06                  1          20
e06                  2          20
e02                  2          10
e02                  3          10
e02                 10          10
e02                 20          10
e03                 30          30
e03                 10          10

ESSN                PNO       HOURS
---------- ---------- ----------
e07                 10          35
e07                 30           5
e04                 30          20
e04                 20          15
e09                 20

16 rows selected.
```

Inserting into dependent table:

SQL>insert into dependent values('e02','Anika','F','05-Apr-1986','Daughter');

SQL>insert into dependent values('e02','arjun','M','25-Oct-1983','Son');

SQL>insert into dependent values('e02','Joy','F','03-May-1958','Spouse');

SQL>insert into dependent values('e04','Arun','M','28-Feb-1942','Spouse');

SQL>insert into dependent values(‘e01’,'Michael','M','04-Jan-1988','Son');

SQL>insert into dependent values('e01','Aruna','F','30-Dec-1988','Daughter');

SQL>insert into dependent values('e01','Elizabeth','F','05-May-1967','Spouse');

Output :

```
SQL> select * from dependent;

ESSN       DEPENDENTNAME                     S BDATE     RELATIONSHIP
---------- --------------------------------- - --------- -------------------------
e02        Anika                             F 05-APR-86 Daughter
e02        arjun                             M 25-OCT-83 Son
e02        Joy                               F 03-MAY-58 Spouse
e04        Arun                              M 28-FEB-42 Spouse
e01        Aruna                             F 30-DEC-88 Daughter
e01        Elizabeth                         F 05-MAY-67 Spouse
```

4. **Perform the following:**

a. Altering a Table, Dropping/Truncating/Renaming Tables

i. Altering a Table

SQL>alter table dept_locations modify dlocation varchar(35);

Output :

```
SQL> desc dept_locations;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DNUMBER                                   NOT NULL NUMBER(38)
 DLOCATION                                 NOT NULL VARCHAR2(25)

SQL> alter table dept_locations modify dlocation varchar(35);
Table altered.
SQL> desc dept_locations
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
                                           NOT NULL NUMBER(38)
                                           NOT NULL VARCHAR2(35)
```

ii. **Truncating**

SQL>truncate table dept_locations;

Output :

```
SQL> truncate table dept_locations;

Table truncated.
```

ii. **Dropping**

SQL>drop table dept_locations;

Output :

```
SQL> drop table dept_locations;

Table dropped.
```

iii. **Renaming**

SQL>rename employee to employee_details;

Output :

```
SQL> rename employee to employee_details;

Table renamed.
```

5. **For a given set of relation schemes, create tables and perform the following Simple Queries, Simple Queries with Aggregate functions, Queries with Aggregate functions (group by and having clause).**

i. **Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary.**

SQL>select sum (salary), max (salary), min (salary), avg (salary) from employee_details

Output :

```
SQL> select sum (salary), max (salary), min (salary), avg (salary)  from employee_details;

SUM(SALARY) MAX(SALARY) MIN(SALARY) AVG(SALARY)
----------- ----------- ----------- -----------
     284000       50000       25000       35500
```

ii. **Retrieve the total number of employees in the company**

SQL>select count (*) from employee_details;

Output :

```
  COUNT(*)
----------
         8
```

iii. **Retrieve the number of employees in the 'Research' department .**

SQL>select count (*) from employee_details, department where dno=dnumber and dname='Research';

Output :

COUNT(*)
4

iv. **For each department, retrieve the department number, the number of employees in the department, and their average salary.**

SQL>select dno, count (*), avg (salary) from employee_details group by dno;

DNO	COUNT(*)	AVG(SALARY)
1	1	38000
5	4	38250
4	3	31000

Find the name of the departments with 2 or more female employees

SQL>select dname from department, employee_details where dnumber = dno and sex = 'F' group by dname having count(ssn) >= 2;

Output :

DNAME
Administration

5. **Execute the fallowing queries**

a. **How the resulting salaries if every employee working on the 'Research' Departments is given a 10% raise.**

SQL>update employee_details set salary=salary+(salary*10/100) where dno = (select dnumber from department where dname='Research');

SQL>select * from employee_details;

Output :

```
SQL> Select * from employee_details;

FNAME           M LNAME           SSN        BDATE     ADDRESS                                                  S     SALARY SUPER_SSN         DNO
--------------- - --------------- ---------- --------- ---------------------------------------------------------- - ---------- ---------- ----------
darshan         B anandh          e01        09-JAN-65 731 Fondren, Houston, TX                                   M      55000 333445555           5
mohan           T ram             e02        08-DEC-55 638 Voss, Houston, TX                                      M      44000 888665555           5
Anisha          J bhatt           e03        19-JAN-68 3321 Castle, Spring, TX                                    F      25000 987654321           4
samantha        S josh            e04        20-JUN-41 291 Berry, Bellaire, TX                                    F      43000 888665555           4
Ramesh          K Narayan         e05        15-SEP-62 975 Fire Oak, Humble, TX                                   M      41800 333445555           5
lisa            A roy             e06        31-JUL-72 5631 Rice ,Houston,TX                                      F      27500 333445555           5
Ahmed           V Jabbar          e07        29-MAR-69 980 Dallas, Houston, TX                                    M      25000 987654321           4
TOM             Z HANKS           e09        14-FEB-71 23 , Springton, TeXas                                      M      38000                     1

8 rows selected.
```

b. **Find the sum of the salaries of all employees of the 'Research' department, as well as the maximum salary, the minimum salary, and the average salary in this department**

SQL>select sum(salary) from employee_details where dno = (select dnumber from department where dname='Research');

SQL>select max(salary) from employee_details where dno = (select dnumber from department where dname='Research');

SQL>select min(salary) from employee_details where dno = (select dnumber from department where dname='Research');

SQL>select avg(salary) from employee_details where dno = (select dnumber from department where dname='Research');

Output :

```
SQL> select sum(salary) from employee_details where dno = (select dnumber from department where dname='Research');

SUM(SALARY)
-----------
     168300

SQL> select max(salary) from employee_details where dno = (select dnumber from department where dname='Research');

MAX(SALARY)
-----------
      55000

SQL> select min(salary) from employee_details where dno = (select dnumber from department where dname='Research');

MIN(SALARY)
-----------
      27500

SQL> select avg(salary) from employee_details where dno = (select dnumber from department where dname='Research');

AVG(SALARY)
-----------
      42075
```

6. **Retrieve the name of each employee Controlled by Department number 5** (use EXISTS operator).

SQL>select fname,lname from employee_details where exists(select pnumber from project where dnum = 5 and pnumber in (select pno from works_on where essn = ssn));

Output :

```
SQL> SELECT fname,lname FROM   employee_details WHERE  EXISTS(SELECT pnumber FROM   project WHERE  dnum
= 5 AND pnumber IN (SELECT pno FROM   works_on WHERE  essn = ssn));

FNAME           LNAME
--------------- ---------------
darshan         anandh
mohan           ram
Ramesh          Narayan
lisa            roy
```

b. **Retrieve the name of each dept and number of employees working in each Department which has at least 2 employees**

SQL>select dname, count(*) from employee_details, department where dno=dnumber group by dname having count(ssn) >= 2;

Output :

```
DNAME                     COUNT(*)
------------------------- ----------
Research                           4
Administration                     3
```

7a. **For each project, retrieve the project number, the project name, and the number of employee who work on that project.(use GROUP BY)**

SQL>select pnumber, pname,count (*) from project, works_on where pnumber = pno group by pnumber, pname;

Output :

```
SQL> SELECT Pnumber, Pname,COUNT (*) FROM PROJECT, WORKS_ON WHERE Pnumber = Pno GROUP BY Pnumber, Pname;


   PNUMBER PNAME            COUNT(*)
---------- -------------- ----------
        20 Reorganization          3
         1 ProductX                2
        10 Computerization         3
        30 Newbenefits             3
         2 ProductY                3
         3 ProductZ                2
```

b. **Retrieve the name of employees who born in the year 1960's**

SQL>select fname from employee_details where bdate between '01-Jan-1960' and '31-Dec-1969';

Output :

```
FNAME
---------------
darshan
Anisha
Ramesh
Ahmed
```

8 . **For each department that has more than two employees, retrieve the department number and the number of its employees who are making more than $20,000.**

SQL>select dname, count (*) from department, employee_details where dnumber=dno and salary>20000 group by dname having count (*) > 2;

Output :

DNAME	COUNT(*)
Research	4
Administration	3

9.**For each project on which more than two employees work, retrieve the project number, project name and the number of employees who work on that project.**

SQL>select pnumber, pname, count (*) from project, works_on where pnumber=pno group by pnumber, pname having count (*) > 2;

Output :

PNUMBER	PNAME	COUNT(*)
2	ProductY	3
30	Newbenefits	3
20	Reorganization	3
10	Computerization	3

10. **For a given set of relation tables perform the following: Creating Views (with and without check option), Dropping views, Selecting from a view**

a . **Creating Views**

i. **SQL**>create view works_on_view as select fname, lname, pname, hours from employee_details, project, works_on where ssn=essn and pno=pnumber;

ii. **SQL**>select * from works_on_view;

Output :

```
SQL> select * from works_on_view;

FNAME           LNAME           PNAME                HOURS
--------------- --------------- --------------- ----------
darshan         anandh          ProductY               7.5
darshan         anandh          ProductX              32.5
mohan           ram             Reorganization          10
mohan           ram             Computerization         10
mohan           ram             ProductZ                10
mohan           ram             ProductY                10
Anisha          bhatt           Computerization         10
Anisha          bhatt           Newbenefits             30
samantha        josh            Reorganization          15
samantha        josh            Newbenefits             20
Ramesh          Narayan         ProductZ                40

FNAME           LNAME           PNAME                HOURS
--------------- --------------- --------------- ----------
lisa            roy             ProductY                20
lisa            roy             ProductX                20
Ahmed           Jabbar          Newbenefits              5
Ahmed           Jabbar          Computerization         35
TOM             HANKS           Reorganization

16 rows selected.
```

i. **SQL**>create view dept_info (dept_name, no_of_emps) as select dname, count (*) from department, employee_details where dnumber=dno group by dname;

ii. **SQL**>select * from dept_info.

Output :

```
DEPT_NAME                    NO_OF_EMPS
-------------------------- ----------
Research                            4
Administration                      3
Headquarters                        1
```

b .**Creating view With Check Option**

SQL>create view dep_view as select * from department where mgrstartdate is not null with check option;

SQL>select * from dep_view ;

Output :

```
SQL> select * from dep_view  ;

DNAME                      DNUMBER MGRSSN     MGRSTARTD
-------------------------- ---------- ---------- ---------
Research                            5 e02        22-MAY-88
Administration                      4 e04        01-JAN-95
Headquarters                        1 e06        19-JUN-81
```

SQL>insert into dep_view values ('Accounts',6, 'e01',null);

Output :

```
SQL> INSERT INTO dep_view values ('Accounts',6, 'e01',null);
INSERT INTO dep_view values ('Accounts',6, 'e01',null)
            *
ERROR at line 1:
ORA-01402: view WITH CHECK OPTION where-clause violation
```

note: In this View if we now try to insert a new row with null value in the mgrstartdate column then it will give an error because the view is created with the condition for mgrstartdate column as NOT NULL.

c . **Selecting from a View**

SQL> select * from dep_view;

Output :

```
SQL> select * from dep_view;

DNAME                          DNUMBER MGRSSN     MGRSTARTD
------------------------------ ---------- ---------- ---------
Research                            5 e02        22-MAY-88
Administration                      4 e04        01-JAN-95
Headquarters                        1 e06        19-JUN-81
```

d. Dropping View

SQL>drop view dep_view;

Output :

```
SQL> DROP VIEW dep_view;

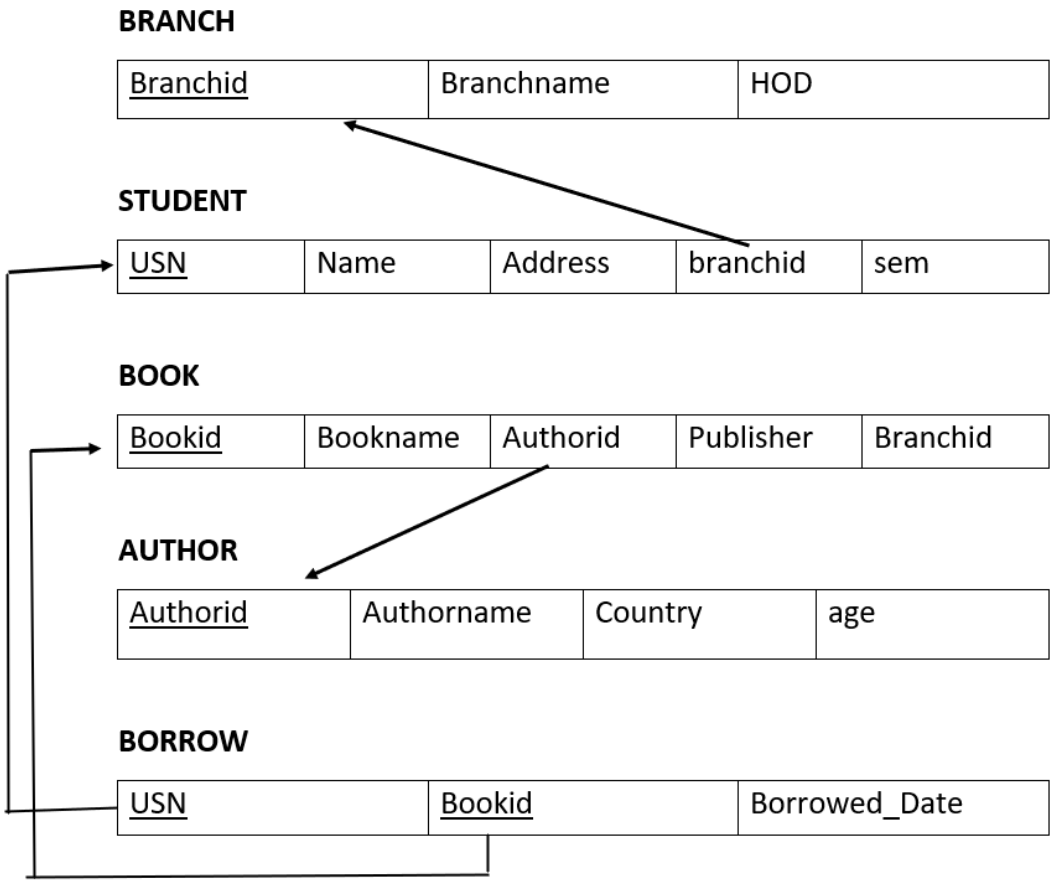
View dropped.
```

PART B

Question no I

I. Create the following tables with properly specifying Primary keys, Foreign keys and solve the following queries.

SCHEMA DIAGRAM:



BRANCH (Branchid, Branchname, HOD)
STUDENT (USN, Name, Address, Branchid, sem)
BOOK (Bookid, Bookname, Authorid, Publisher, Branchid)
AUTHOR (Authorid, Authorname, Country, age)
BORROW (USN, Bookid, Borrowed_Date)

2. **Perform the following:**
   a. **Creating Tables (With and Without Constraints), Inserting/Updating/Deleting Records in a Table, Saving (Commit) and Undoing(rollback)**
   Execute the following Queries:

SQL> create table branch (Branchid number(10) primary key, BranchName varchar(5), Hod varchar(10));

```
SQL> desc branch;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 BRANCHID                                  NOT NULL NUMBER(10)
 BRANCHNAME                                         VARCHAR2(5)
 HOD                                                VARCHAR2(10)
```

SQL>create table student (USN number(10) primary key, Name varchar(10), Address varchar(20), Branchid number(10) references branch,Sem varchar(5));

```
SQL> desc student;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 USN                                       NOT NULL NUMBER(10)
 NAME                                               VARCHAR2(10)
 ADDRESS                                            VARCHAR2(20)
 BRANCHID                                           NUMBER(10)
 SEM                                                VARCHAR2(5)
```

SQL>create table author (Authorid number(10) primary key, authorrname varchar2(15), country varchar(10), age number(5));

```
SQL> desc author;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 AUTHORID                                  NOT NULL NUMBER(10)
 AUTHORRNAME                                        VARCHAR2(15)
 COUNTRY                                            VARCHAR2(10)
 AGE                                                NUMBER(5)
```

SQL>create table book (Bookid number(5) primary key, Bookname varchar2(10), authorid number(10) references author, publisher varchar(10), branchid number(10) references branch);

```
SQL> desc book;
 Name                                      Null?    Type
 ----------------------------------------- -------- ------------------
 BOOKID                                    NOT NULL NUMBER(5)
 BOOKNAME                                           VARCHAR2(10)
 AUTHORID                                           NUMBER(10)
 PUBLISHER                                          VARCHAR2(10)
 BRANCHID                                           NUMBER(10)
```

SQL>create table borrow (USN number(10) references student, Bookid number(10) references book, Borrowed_date date);

```
SQL> desc borrow;
 Name                                      Null?    Type
 ----------------------------------------- -------- ------------------
 USN                                                NUMBER(10)
 BOOKID                                             NUMBER(10)
 BORROWED_DATE                                      DATE
```

SQL> insert into branch values('101','BCA','Prof.Tilak');
SQL>insert into branch values('102','BBA','Dr.Alok');
SQL>insert into branch values('103','BCom','Prof.Sunil');

```
  BRANCHID BRANC HOD
---------- ----- ----------
       101 BCA   Prof.Tilak
       102 BBA   Dr.Alok
       103 BCom  Prof.Sunil
```

SQL> insert into student values('1001','Aditya','UP','101','2');
SQL> insert into student values('1002','Aakash','Bidar','102','1');
SQL> insert into student values('1003','Nagaseshu','Bangalore','103','3');
SQL> insert into student values(1004,'Lakshmi','Nepal',103,'4');

```
SQL> select * from student;

       USN NAME       ADDRESS                 BRANCHID SEM
---------- ---------- -------------------- ---------- -----
      1001 Aditya     UP                          101 2
      1002 Aakash     Bidar                       102 1
      1003 Nagaseshu  Bangalore                   103 3
      1004 Lakshmi    Nepal                       103 4
```

SQL> insert into author values('001','Vicky','India','26');
SQL> insert into author values('002','Suman','Nepal','59');
SQL> insert into author values('003','John','America','89');

```
SQL> select * from author;

 AUTHORID AUTHORRNAME     COUNTRY          AGE
--------- --------------- ---------- ----------
        1 Vicky           India              26
        2 Suman           Nepal              59
        3 John            America            89
```

SQL> insert into book values('10','Cloud','1','Penguin','101');
SQL> insert into book values('11','DBMS','2','Pearson','102');
SQL> insert into book values('12','Forest','3','Springer','103')

```
SQL> select * from book;

    BOOKID BOOKNAME     AUTHORID PUBLISHER    BRANCHID
---------- ---------- ---------- ---------- ----------
        10 Cloud               1 Penguin           101
        11 DBMS                2 Pearson           102
        12 Forest              3 Springer          103
```

SQL> insert into borrow values('1001','10','02-feb-2012');
SQL> insert into borrow values('1002','11','22-Mar-2017');
SQL> insert into borrow values('1003','12','30-dec-2022');
SQL>insert into borrow values('1001','12','03-Mar-2020');
SQL>insert into borrow values('1003','11','05-Jun-2019');

```
SQL> select * from borrow;

       USN     BOOKID BORROWED_
---------- ---------- ---------
      1001         10 02-FEB-12
      1002         11 22-MAR-17
      1001         12 03-MAR-20
      1003         11 05-JUN-19
```

3. a. **List the details of Students who are all studying in 2nd sem BCA.**

SQL>select * from student where sem=’2’ and branchid in (select branchid from branch where branchname='BCA');

```
SQL> select * from student where sem='2' and branchid in (select branchid from branch where branchname='BCA');

     USN NAME      ADDRESS              BRANCHID SEM
--------- --------- -------------------- --------- -----
    1001 Aditya    UP                         101 2
```

b. List the students who are not borrowed any books.

SQL>select * from student where usn not in (select usn from borrow);

```
SQL> select * from student where usn not in (select usn from borrow);

       USN NAME       ADDRESS                BRANCHID SEM
---------- ---------- -------------------- ---------- -----
      1004 Lakshmi    Nepal                       103 4
```

2. a .**Display the USN, Student name, Branch_name, Book_name, Author_name, Books_Borrowed_ Date of 2nd sem BCA Students who borrowed books.**

SQL>Select student.usn,student.name,branch.branchname,book.bookname,author.authorrname,borrowed_date from student,branch,book,author, borrow where student.usn=borrow.usn and borrow.bookid=book.bookid and book.authorid=author.authorid and student.branchid=branch.branchid and student.sem='2' and branch.branchname='BCA';

```
       USN NAME       BRANC BOOKNAME   AUTHORRNAME     BORROWED_
---------- ---------- ----- ---------- --------------- ---------
      1001 Aditya     BCA   Cloud      Vicky           02-FEB-12
      1001 Aditya     BCA   Forest     John            03-MAR-20
```

b. Display the number of books written by each author.

SQL>select count(*) as “Total” , authorid from book group by authorid;

```
SQL> select count(*) as "Total" , authorid from book group by authorid;

     Total   AUTHORID
---------- ----------
         1          1
         1          2
         1          3
```

4**. a. Display the student details who borrowed more than two books.**

SQL > select * from student where usn in (select usn from borrow

group by usn having count(usn) >=2);

```
SQL> select * from student where usn in ( select usn from borrow
  2  group by usn having count(usn) >=2);

       USN NAME       ADDRESS                BRANCHID SEM
---------- ---------- -------------------- ---------- -----
      1001 Aditya     UP                          101 2
```

b. Display the student details who borrowed books of more than one author.

SQL>select * from student s where exists (select br.usn from borrow br

join book bk on br.bookid=bk.bookid where br.usn=s.usn group by usn

having count(distinct authorid)>1);

```
SQL> select * from student s where exists (select br.usn from borrow br
  2  join book bk on br.bookid=bk.bookid where br.usn=s.usn group by usn
  3  having count(distinct authorid)>1);

       USN NAME       ADDRESS                BRANCHID SEM
---------- ---------- -------------------- ---------- -----
      1001 Aditya     UP                          101 2
```

5. **Display the Book names in descending order of their names.**

SQL> select bookname from book order by bookname desc;

```
SQL> select bookname from book order by bookname desc;

BOOKNAME
----------
Forest
DBMS
Cloud
```

PART – B Question no II

II. Consider the following schema:

STUDENT(USN,name,DOB,branch,mark1,mark2,mark3,total,GPA)

I. Perform the following:

Creating Tables (With and Without Constraints) inserting / Updating /Deleting/Records in a Table ,Saving(Commit) and Undoing(rollback)

1.Creating table

SQL>create table student(Usn number(10) primary key, Name varchar2(20), Dob date, Branch varchar2(20), Mark1 number(10), Mark2 number(10), Mark3 number(10), Total number(10), Gpa number(10));

Displaying The attributes:

SQL>desc student;

```
SQL> desc student;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 USN                                       NOT NULL NUMBER(10)
 NAME                                               VARCHAR2(20)
 DOB                                                DATE
 BRANCH                                             VARCHAR2(20)
 MARK1                                              NUMBER(10)
 MARK2                                              NUMBER(10)
 MARK3                                              NUMBER(10)
 TOTAL                                              NUMBER(10)
 GPA                                                NUMBER(10)
```

2.Inserting tuples (insert at least 5 tuples with different branch)

SQL> insert into student(Usn,Name,Dob,Branch,Mark1,Mark2,Mark3)values(‘101’,’aravind’,

‘26/jan/2003’,’BCA’,’90’,’85’,’99’);

SQL> insert into student(Usn,Name,Dob,Branch,Mark1,Mark2,Mark3)values(‘102’,’buvan’,

‘26/mar/2002’,’BCA’,’95’,’78’,’89’);

SQL> insert into student(Usn,Name,Dob,Branch,Mark1,Mark2,Mark3)values(‘103’,shankar’,

‘24/feb/2000’,’BBA’,’90’,’91’,’93’);

SQL> insert into student(Usn,Name,Dob,Branch,Mark1,Mark2,Mark3)values('104',sachin',

'12/sep/2000','BBA','96','94','97');

SQL> insert into student(Usn,Name,Dob,Branch,Mark1,Mark2,Mark3)values('105','utkarsh',

'12/sep/2003','BCOM','96','95','92');

SQL>select * from student;

```
SQL> select * from student;

USN NAME                 DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL        GPA
--- -------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
101 aravind              26-JAN-03 BCA                       90         85         99
102 buvan                26-MAR-02 BCA                       95         78         89
103 shankar              24-FEB-00 BBA                       90         91         93
104 sachin               12-SEP-00 BBA                       96         94         97
105 utkash               12-SEP-03 BCOM                      96         95         92
```

3.Updating Total marks and Gpa

SQL>update student set Total=Mark1+Mark2+Mark3;

SQL>update student set Gpa=Total/3;

Displaying the table after updating

SQL>select *from student;

```
SQL>
SQL> select * from student;

USN NAME                 DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL        GPA
--- -------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
101 aravind              26-JAN-03 BCA                       90         85         99        274         91
102 buvan                26-MAR-02 BCA                       95         78         89        262         87
103 shankar              24-FEB-00 BBA                       90         91         93        274         91
104 sachin               12-SEP-00 BBA                       96         94         97        287         96
105 utkash               12-SEP-03 BCOM                      96         95         92        283         94
```

Displaying a tuple from the table

SQL> delete from student where usn='103'; (3rd row will be deleted)

Displaying the table after deletion

```
SQL> select * from student;

      USN NAME                DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL        GPA
--------- -------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
      101 aravind              26-JAN-03 BCA                       90         85         99        274         91
      102 buvan                26-MAR-02 BCA                       95         78         89        262         87
      104 sachin               12-SEP-00 BBA                       96         94         97        287         96
      105 utkash               12-SEP-03 BCOM                      96         95         92        283         94
```

5. Saving(Commit)and Undoing(rollback) operations

SQL>commit; (To Save)

SQL>rollback; (To Undo)

```
SQL> commit;

Commit complete.

SQL> select * from student;

      USN NAME                DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL        GPA
--------- -------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
      101 aravind              26-JAN-03 BCA                       90         85         99        274         91
      102 buvan                26-MAR-02 BCA                       95         78         89        262         87
      104 sachin               12-SEP-00 BBA                       96         94         97        287         96
      105 utkash               12-SEP-03 BCOM                      96         95         92        283         94

SQL> delete from student where Usn='102';

1 row deleted.

SQL> select * from student;

      USN NAME                DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL        GPA
--------- -------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
      101 aravind              26-JAN-03 BCA                       90         85         99        274         91
      104 sachin               12-SEP-00 BBA                       96         94         97        287         96
      105 utkash               12-SEP-03 BCOM                      96         95         92        283         94

SQL> rollback;

Rollback complete.

SQL> select * from student;

      USN NAME                DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL        GPA
--------- -------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
      101 aravind              26-JAN-03 BCA                       90         85         99        274         91
      102 buvan                26-MAR-02 BCA                       95         78         89        262         87
      104 sachin               12-SEP-00 BBA                       96         94         97        287         96
      105 utkash               12-SEP-03 BCOM                      96         95         92        283         94
```

6.Execute the following queries:

c. Find the GPA score of all the students.

SQL>**select sum(Gpa) as "Tot" from student;**

```
SQL> select sum(Gpa) as "Tot" from student;

       Tot
----------
       368
```

d. Find the students who were born on a particular year of birth from the DOB Column.

SQL>select *from student where Dob>'26-jan-2000' order by Dob asc ;

```
SQL> select *from student where Dob>'26-jan-2000' order by Dob  asc ;

      USN NAME               DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL        GPA
---------- ------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
      104 sachin             12-SEP-00 BBA                          96         94         97        287         96
      102 buvan              26-MAR-02 BCA                          95         78         89        262         87
      101 aravind            26-JAN-03 BCA                          90         85         99        274         91
      105 utkash             12-SEP-03 BCOM                         96         95         92        283         94
```

7 a. List the students who are studying in a particular branch of study.

SQL>select * from student where branch='BCA';

```
SQL> select * from  student  where  branch='BCA';

      USN NAME               DOB       BRANCH                MARK1      MARK2      MARK3      TOTAL    GPA
---------- ------------------- --------- -------------------- ---------- ---------- ---------- ---------- ----------
      101 aravind            26-JAN-03 BCA                          90         85         99        274    91
      102 buvan              26-MAR-02 BCA                          95         78         89        262    87
```

b. Find the maximum GPA score of the student branch-wise.

SQL>select max(Gpa)from student where branch='BCA';

```
SQL>
SQL> select max(Gpa)from student where branch='BCA';

  MAX(GPA)
----------
        91
```